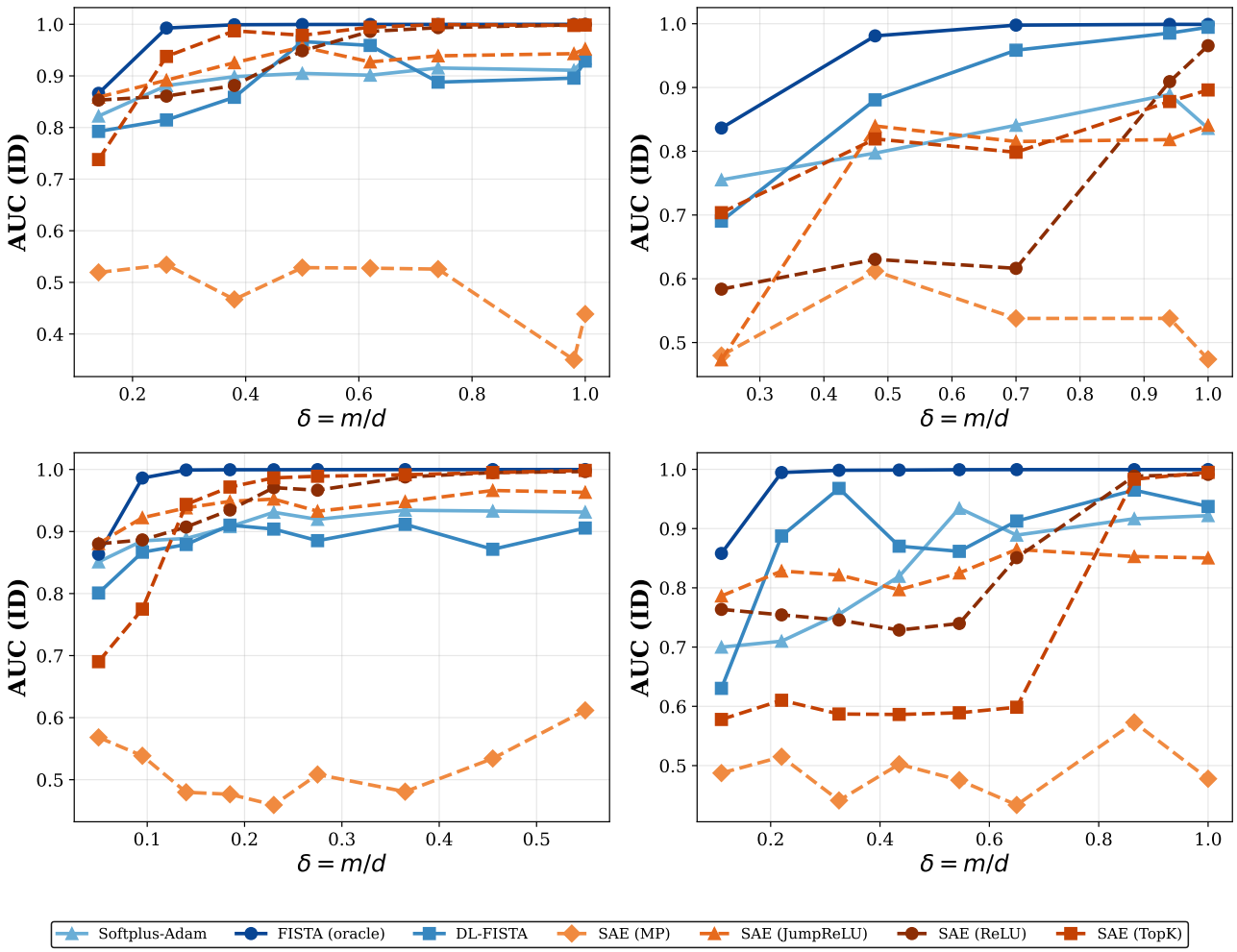

Figure 61: **Phase transition: AUC (ID).** ID AUC is high for most methods, but the gap between per-sample and amortised methods remains visible even in-distribution.

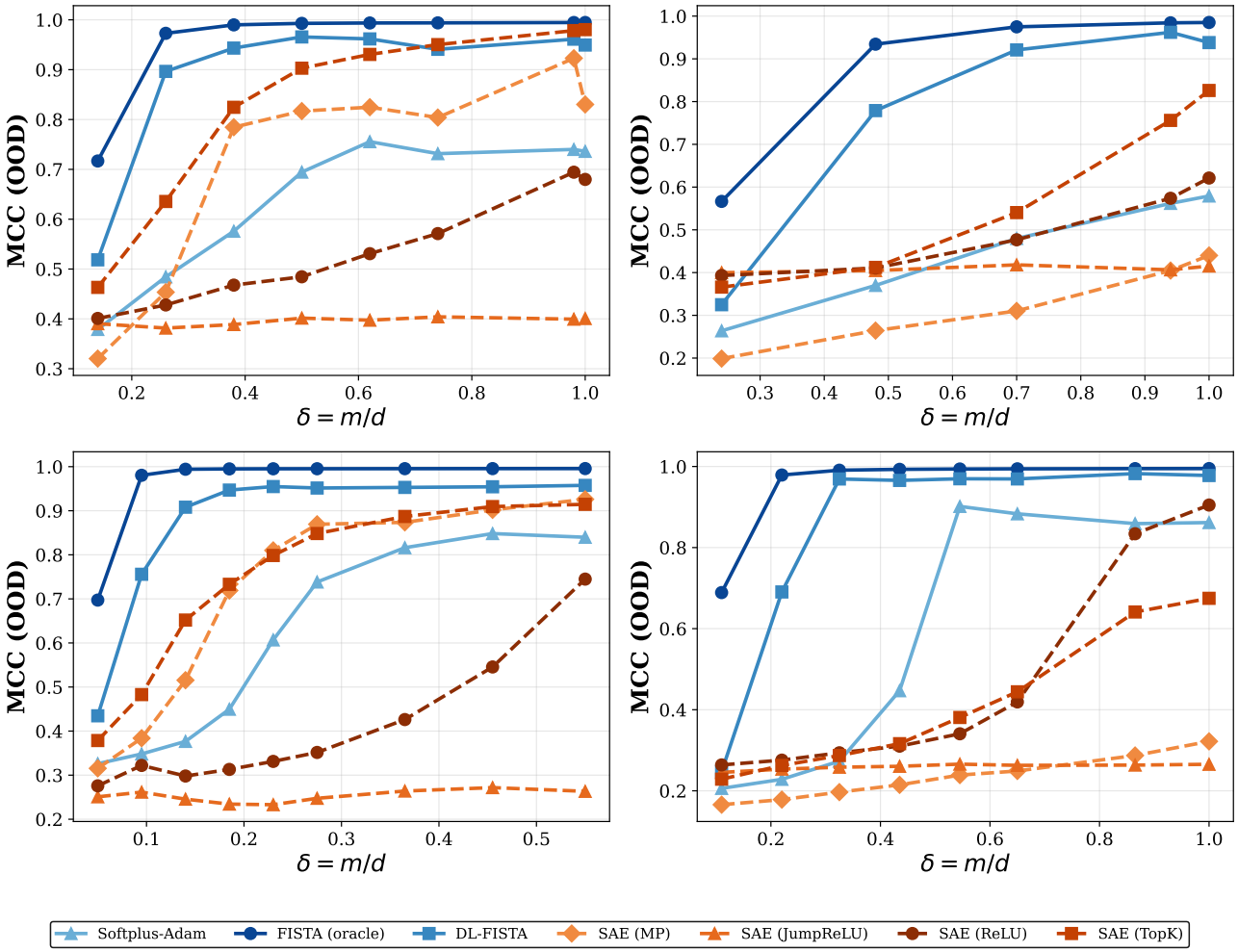

Figure 62: **Phase transition: MCC (OOD).** OOD MCC shows the clearest separation: per-sample methods recover the full latent structure under novel compositions while SAEs fail to identify latents OOD.

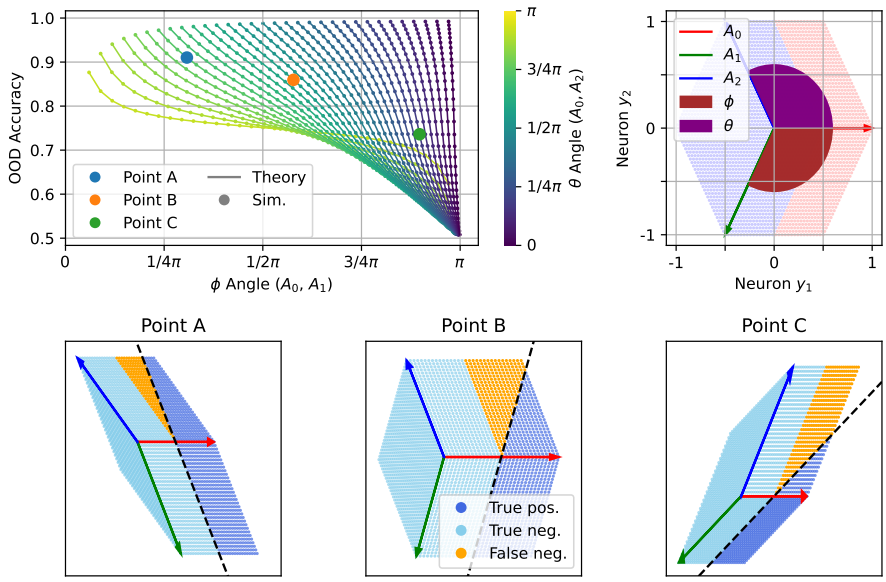

Figure 63: **Theory. Top left,** shows the theoretically predicted accuracy and simulations of a perfect linear classifier, trained and tested (OOD) on distinct latent combinations (see Figure 1). **Top right,** illustrates the geometry of the classification problem (red and blue classes) with the directions of the decoder A columns for each latent and the angles (ϕ, θ) between them. **Bottom,** shows the resulting geometry for three sample points from the first plot.